

COMPASS Synoptic CB Porewater: DOC

May 2024 Samples

2025-12-18

##Setup - Change things here & write any notes

```
#identify section
cat("Setup Information")
```

Setup Information

Run information - PLEASE CHANGE

```
Date_Run = "6/03/24" #Date that instrument was run
Run_by = "Stephanie J. Wilson" #Instrument user
Script_run_by = "Stephanie J. Wilson" #Code user
run_notes = "Samples were put in the wrong vials, so
MSM_202405_UP_LysA_ was switched to MSM_202405_UP_LysB_,
MSM_202405_UP_LysB_ was switched to MSM_202405_UP_LysC_,
MSM_202405_UP_LysC_ was switched to MSM_202405_UP_LysA_.
Second NPOC Curve did not look good. One duplicate out of range and some checks
below expected concentration, but not by much - run okay." #any notes from the run
samples <- c("GCW", "GWI", "MSM", "SWH") #whatever identifies your samples within the same names
samples_pattern <- paste(samples, collapse = "|")
#samples_pattern <- "GCW" #use this instead of the line above if you have only one site code
chks_name = "Chk_Std_50mgC_2mgN" #what did you name your check standards?
```

File Names - PLEASE CHANGE

#file path and name for raw summary data file

```
#raw_file_name = "Raw Data/COMPASS_SynopticCB_PW_DOC_202505.txt" #example
raw_file_name = "Raw Data/COMPASS_SynopticCB_PW_DOC_202405.txt"
```

#file path and name for raw all peaks file

```
#raw_allpeaks_name = "Raw Data/COMPASS_SynopticCB_PW_DOC_202505_allpeaks.txt" #example
raw_allpeaks_name = "Raw Data/COMPASS_SynopticCB_PW_DOC_202405_allpeaks.txt"
```

#file path and name of processed data file

```
#processed_file_name = "Processed Data/COMPASS_SynopticCB_PW_Processed_DOC_202505.csv" #example
processed_file_name = "Processed Data/COMPASS_SynopticCB_PW_Processed_DOC_202405.csv" #example
```

Log Files - PLEASE CHECK

#downloaded metadata csv - downloaded from Google drive as csv for this year

```
Raw_Metadata = "Raw Data/COMPASS_SynopticCB_PW_SampleLog_2024.csv"
```

#qaqc log file path for this year

```
Log_path = "Raw Data/COMPASS_Synoptic_TOCTN_QAQClog_2024.csv"
```

```
##Set Up Code
```

```
##Read in metadata and create similar sample IDs for matching to samples
```

Import Data Functions

Import Sample Data

```
## Import Sample Data
```

```
## New names:
```

```
## * ' ' -> '...14'
```

```
## # A tibble: 6 x 4
```

```
##   sample_name      npoc_raw tdn_raw run_datetime
##   <chr>          <dbl>   <dbl> <chr>
## 1 202405_SWH_UP_LysA_10cm    14.5   0.373 5/31/2024 9:08:46 PM
## 2 202405_SWH_UP_LysA_20cm    17.5   0.525 5/31/2024 9:33:40 PM
## 3 202405_SWH_UP_LysA_45cm    13.6   0.395 5/31/2024 9:58:11 PM
## 4 202405_SWH_UP_LysB_10cm    29.9   0.626 5/31/2024 10:23:37 PM
## 5 202405_SWH_UP_LysB_20cm    52.7   1.33  5/31/2024 10:49:45 PM
## 6 202405_SWH_UP_LysC_10cm    27     0.616 5/31/2024 11:06:37 PM
```

#Note from Zoe and Steph: It looks like we did not get any samples for MSM_202405_UP_LysA_10cm. According to the corrections on the scanned datasheets, it looks like for Upland, B samples were accidentally put in the A vials, C samples were put in B vials, and A samples were put in C vials. So, I'm thinking I can check if any analytes have MSM_202405_UP_LysA_10cm, then all the samples need to be corrected A ->B, B -> C and C->A.

```
## [1] "The string '202405_MSM_UP_LysA_10cm' was found in the dataframe."
```

Assessing Standard Curves

```
## Assess the Standard Curves
```

```
## New names:
```

```
## * ' ' -> '...18'
```

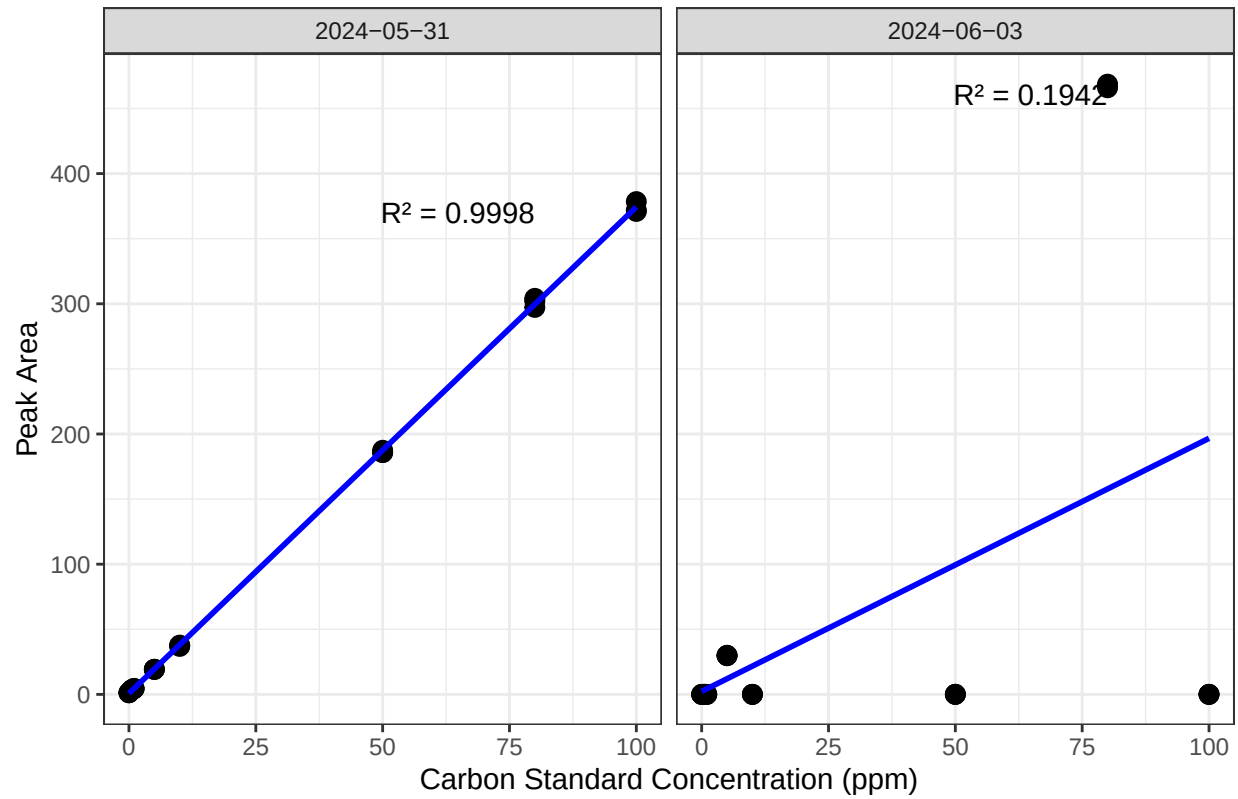
```
## Warning: One or more parsing issues, call 'problems()' on your data frame for details,
## e.g.:
```

```
##   dat <- vroom(...)
```

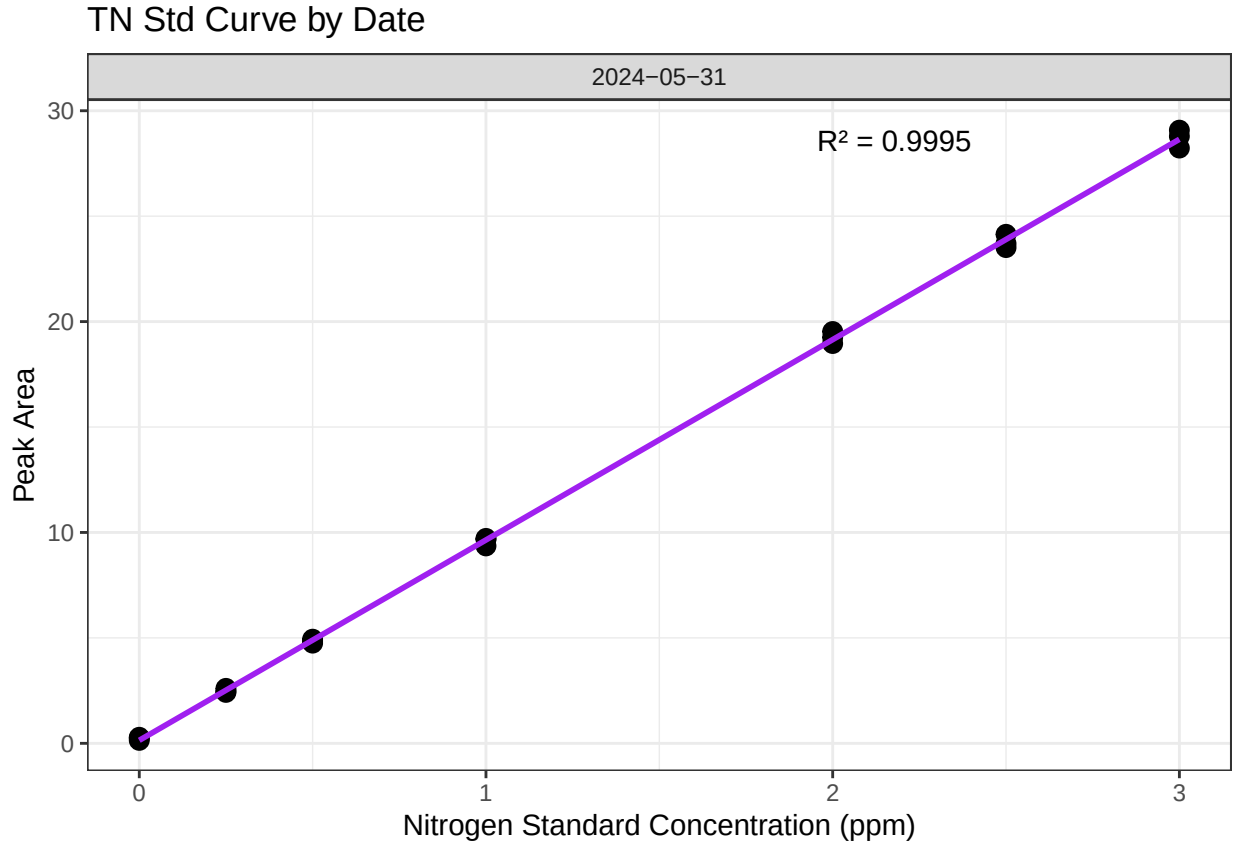
```
##   problems(dat)
```

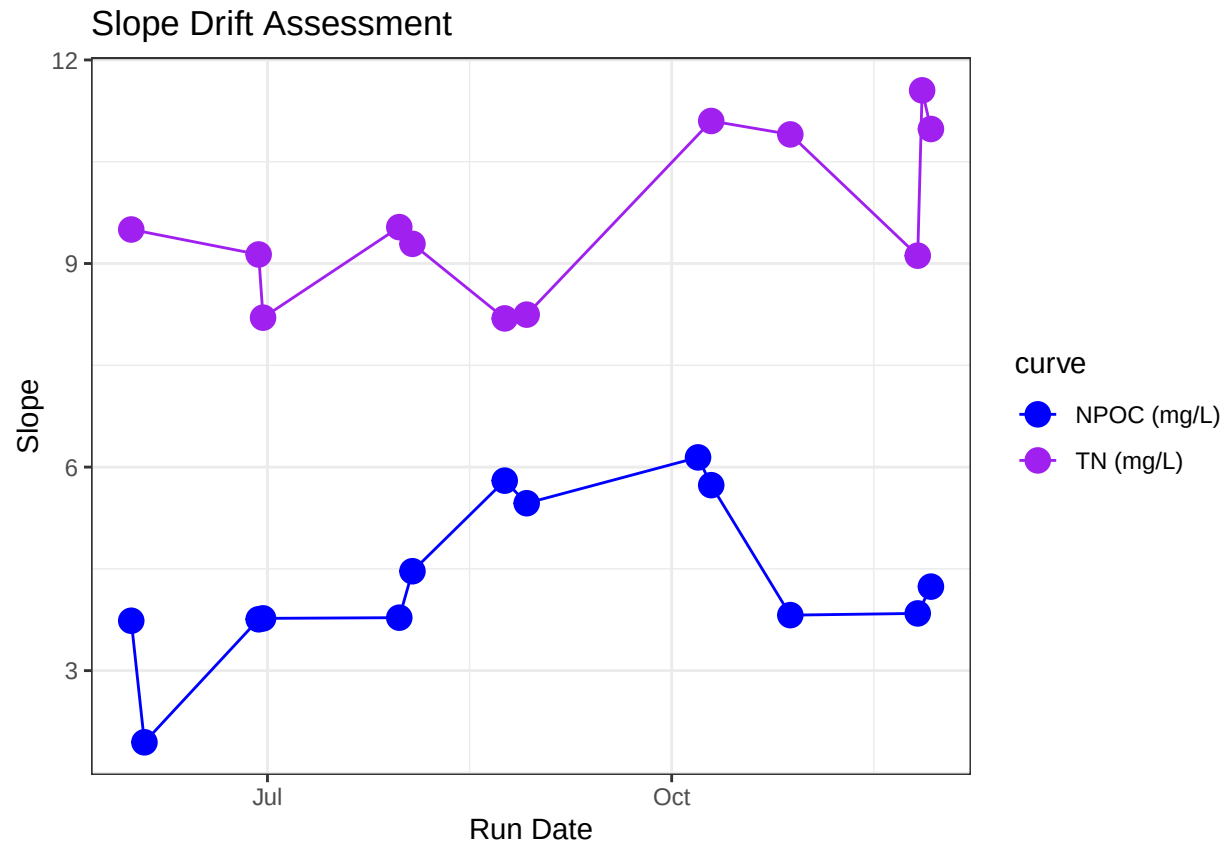
```
## 'geom_smooth()' using formula = 'y ~ x'
```

NPOC Std Curve by Date



```
## 'geom_smooth()' using formula = 'y ~ x'
```





```
## [1] "NPOC Curve r2 GOOD"
```

```
## [1] "TN Curve r2 GOOD"
```

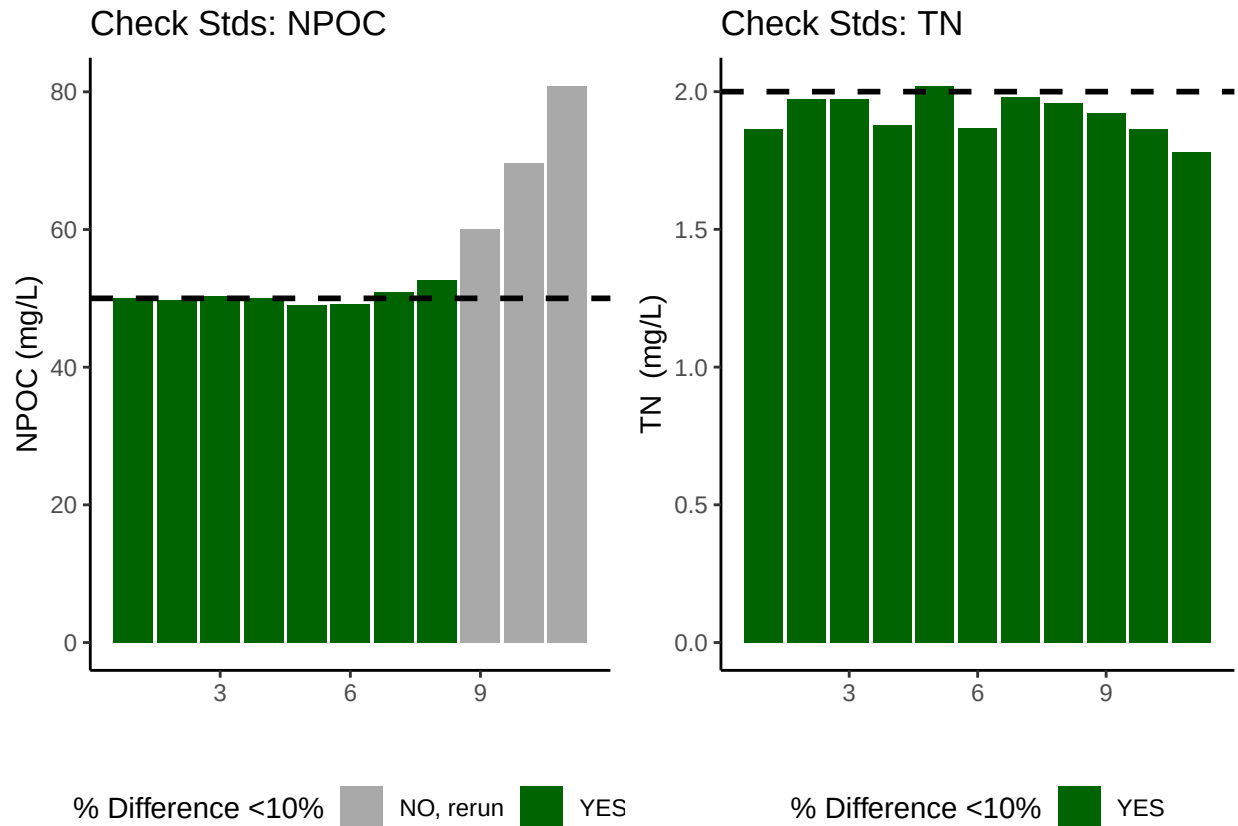
Assess Check Standards

```
## Assess the Check Standards
```

```
## New names:
## * ' ' -> '...14'
```

```
## [1] "Carbon CHECK STANDARD RSD TOO HIGH - REASSESS"
```

```
## [1] "Nitrogen Check Standard RSD within Range"
```



```
## [1] ">60% of Carbon Check Standards are within range of expected concentration"
```

```
## [1] ">60% of Nitrogen Check Standards are within range of expected concentration"
```

Assess Blanks

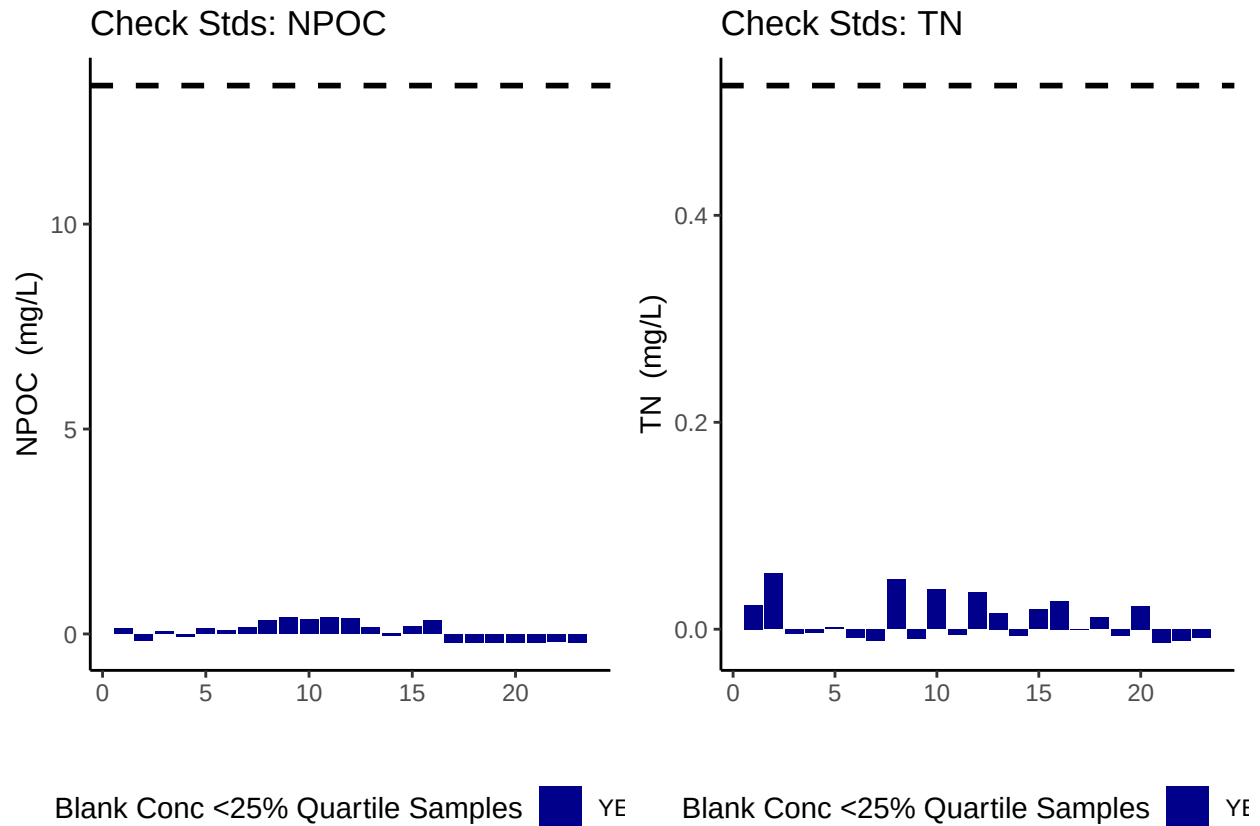
```
## Assess Blanks
```

```
## New names:
```

```
## * ' ' -> '...14'
```

```
## [1] ">60% of Carbon Blank concentrations are lower 25% quartile of samples"
```

```
## [1] ">60% of Nitrogen Blank concentrations are lower 25% quartile of samples"
```



```
## carbon blanks:
```

```
## [1] 0.06258217
```

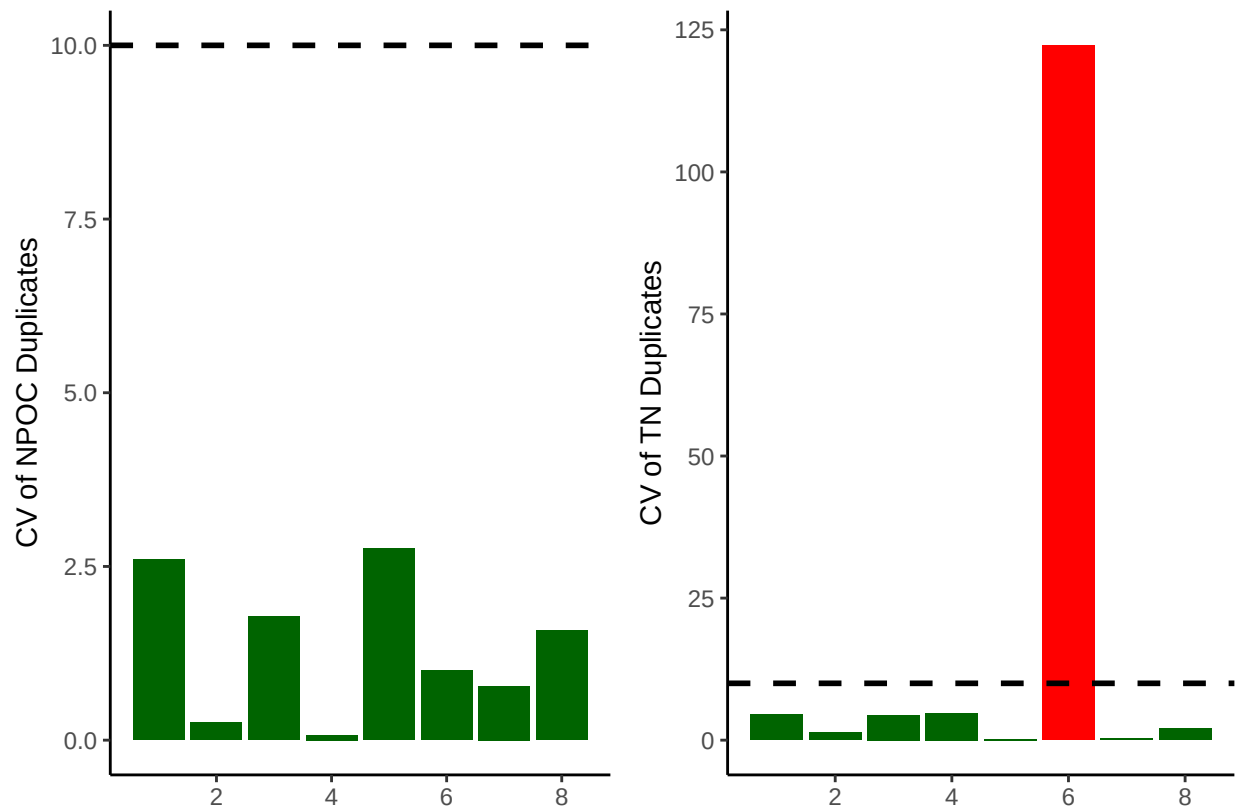
```
## nitrogen blanks:
```

```
## [1] 0.009127391
```

Assess Duplicates

```
## Assess Duplicates
```

```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

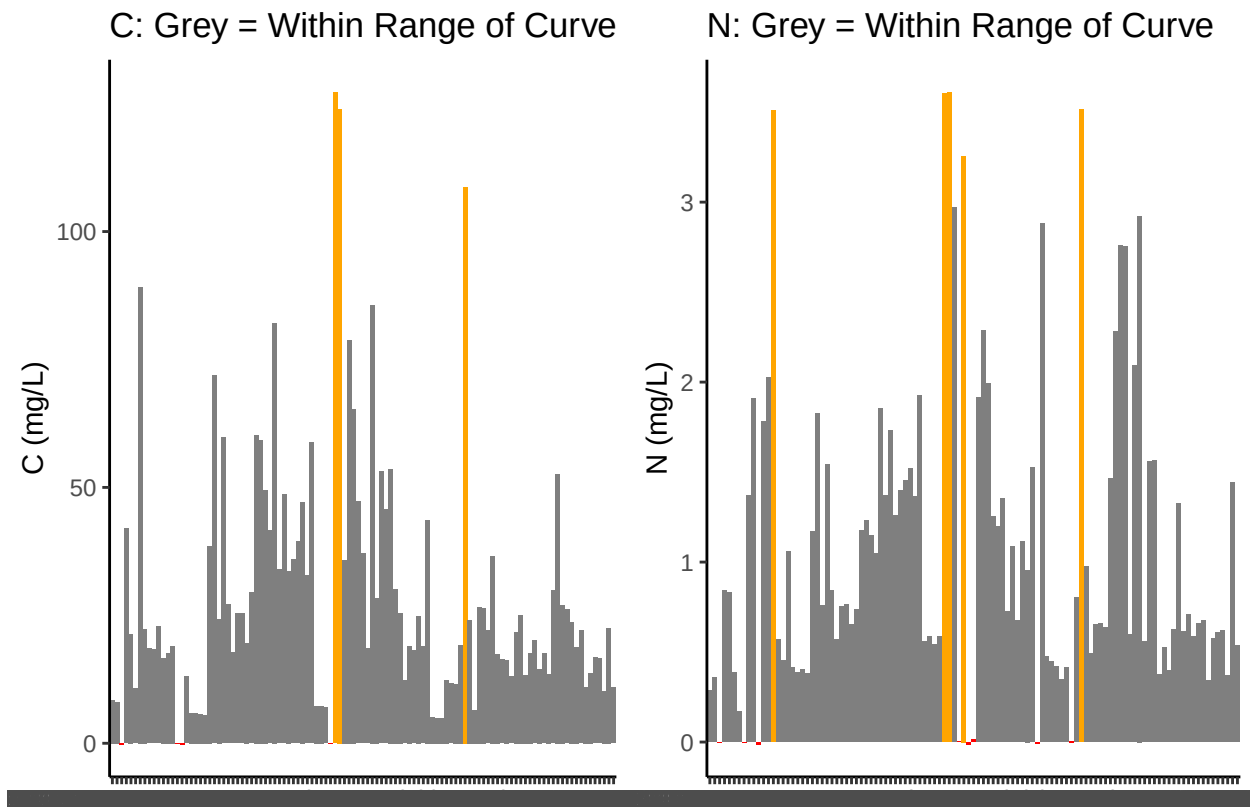


```
## [1] ">60% of Carbon Duplicates have a CV <10%"
```

```
## [1] ">60% of Nitrogen Duplicates have a CV <10%"
```

Sample Flagging - Are samples Within the range of the curve?

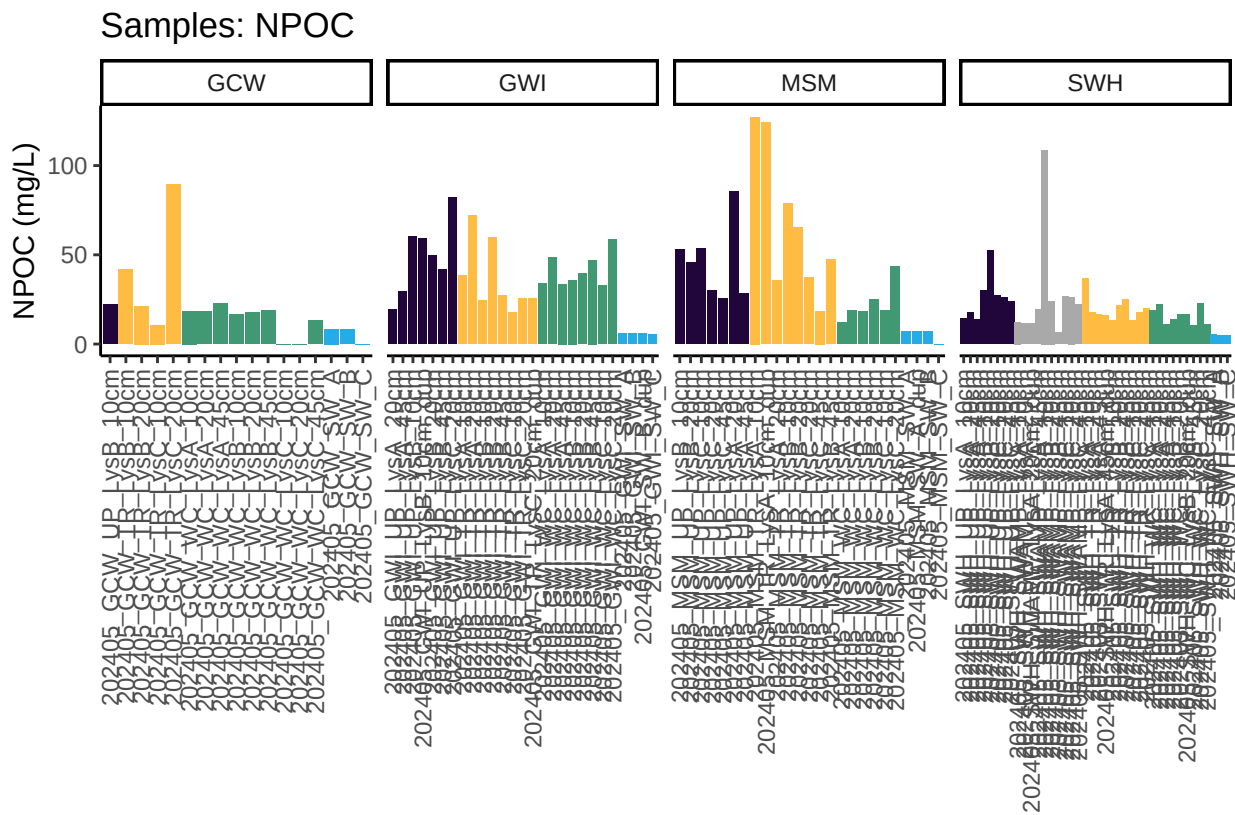
```
## Sample Flagging
```

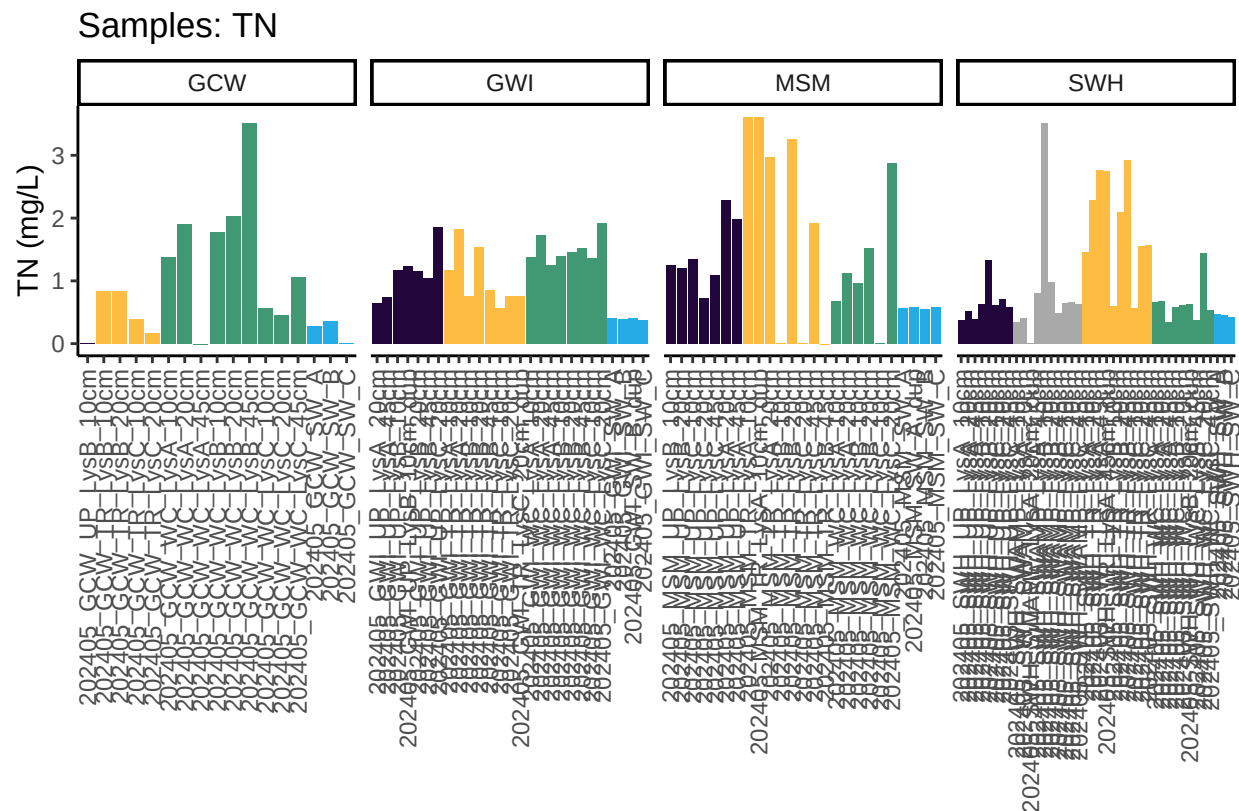



Visualize Data by Plot

```
## Visualize Data
```

```
## Warning in rbind(c("202405", "SWH", "UP", "LysA", "10cm"), c("202405", "SWH", :  
## number of columns of result is not a multiple of vector length (arg 1)
```





Convert data from mg/L to uMoles/L

Check to see if samples run match metadata & merge info

```
## Check Sample IDs with Metadata
```

```
## All sample IDs are present in metadata.
```

Export Processed Data

```
## Export Processed Data
```

```
## # A tibble: 6 x 21
```

```
##   Project      Region Site Zone Replicate Depth_cm Sample_ID Year Month Day
##   <chr>        <chr> <chr> <fct> <chr>      <int> <chr>      <int> <int> <int>
## 1 COMPASS: Sy~ CB    SWH  UP    A          10 202405_S~  2024    5   15
## 2 COMPASS: Sy~ CB    SWH  UP    A          20 202405_S~  2024    5   15
## 3 COMPASS: Sy~ CB    SWH  UP    A          45 202405_S~  2024    5   15
## 4 COMPASS: Sy~ CB    SWH  UP    B          10 202405_S~  2024    5   15
## 5 COMPASS: Sy~ CB    SWH  UP    B          20 202405_S~  2024    5   15
## 6 COMPASS: Sy~ CB    SWH  UP    C          10 202405_S~  2024    5   15
## # i 11 more variables: Time <chr>, Time_Zone <chr>, npoc_mgL <dbl>,
## #   npoc_uM <dbl>, npoc_flag <chr>, tdn_mgL <dbl>, tdn_uM <dbl>,
## #   tdn_flag <chr>, Analysis_runtime <chr>, Run_notes <chr>, Field_notes <chr>
```

#end